

Zoho Voice API Integration Guide for Laravel

This guide outlines the steps to integrate Zoho Voice API into a Laravel project, covering authentication, service setup, and example API calls.

1. Prerequisites

Before you begin, ensure you have the following:

- A **Zoho Voice account**.
- A **Laravel project** set up and running.
- Basic understanding of **Laravel development** and **OAuth 2.0**.

2. Zoho OAuth 2.0 Setup

Zoho Voice APIs use OAuth 2.0 for authentication. You will need to register your application with Zoho to obtain the necessary credentials.

2.1 Register Your Application with Zoho

1. Go to the [Zoho API Console](#).
2. Click on **GET STARTED** or **Add Client**.
3. Choose the **Client Type** that best suits your Laravel application (e.g., "Web-based Application" for a typical server-side Laravel app).
4. Fill in the required details:
 - **Client Name**: A descriptive name for your application.
 - **Homepage URL**: The base URL of your Laravel application.
 - **Authorized Redirect URIs**: This is crucial. It should be the URL where Zoho will redirect the user after they grant permission to your application. For Laravel, this will typically be a route in your application, e.g., `https://your-laravel-app.com/zoho/oauth/callback`.
5. After registration, you will receive a **Client ID** and **Client Secret**. Keep these secure.

2.2 Configure Laravel for OAuth Credentials

Add your Zoho Client ID, Client Secret, and Redirect URI to your Laravel project's `.env` file:

Plain Text

```
ZOHO_CLIENT_ID=your_zoho_client_id
ZOHO_CLIENT_SECRET=your_zoho_client_secret
ZOHO_REDIRECT_URI=https://your-laravel-app.com/zoho/oauth/callback
ZOHO_ACCOUNTS_URL=https://accounts.zoho.com
ZOHO_VOICE_API_URL=https://voice.zoho.com/rest/json
```

Create a configuration file (e.g., `config/zoho.php`) to easily access these values:

PHP

```
<?php

return [
    'client_id' => env('ZOHO_CLIENT_ID'),
    'client_secret' => env('ZOHO_CLIENT_SECRET'),
    'redirect_uri' => env('ZOHO_REDIRECT_URI'),
    'accounts_url' => env('ZOHO_ACCOUNTS_URL'),
    'voice_api_url' => env('ZOHO_VOICE_API_URL'),
    'scopes' => [
        'ZohoVoice.sms.CREATE',
        'ZohoVoice.sms.READ',
        'ZohoVoice.users.READ',
        'ZohoVoice.calllogs.READ'
    ], // Add all necessary scopes here
];
```

3. OAuth Workflow Implementation in Laravel

This section details how to implement the authorization code flow to obtain and manage access tokens.

3.1 Install Guzzle HTTP Client

Laravel's HTTP client is powered by Guzzle, which is suitable for making API requests. Ensure it's installed:

Bash

```
composer require guzzlehttp/guzzle
```

3.2 Create an OAuth Controller

Create a controller to handle the OAuth redirection and token exchange:

Bash

```
php artisan make:controller ZohoOAuthController
```

PHP

```
<?php
```

```
namespace App\Http\Controllers;
```

```
use Illuminate\Http\Request;
```

```
use Illuminate\Support\Facades\Http;
```

```
use Illuminate\Support\Facades\Session;
```

```
class ZohoOAuthController extends Controller
```

```
{
```

```
    public function redirectToZoho( )
```

```
    {
```

```
        $clientId = config('zoho.client_id');
```

```
        $redirectUri = config('zoho.redirect_uri');
```

```
        $accountsUrl = config('zoho.accounts_url');
```

```
        $scopes = implode(',', config('zoho.scopes'));
```

```
        $authUrl = "{$accountsUrl}/oauth/v2/auth?scope={$scopes}&client_id={$clientId}";
```

```
        return redirect($authUrl);
```

```
    }
```

```
    public function handleZohoCallback(Request $request)
```

```
    {
```

```
        $code = $request->query('code');
```

```
        if (!$code) {
```

```
            return redirect('/')->with('error', 'Authorization code not received');
        }
```

```
        $clientId = config('zoho.client_id');
```

```
        $clientSecret = config('zoho.client_secret');
```

```
        $redirectUri = config('zoho.redirect_uri');
```

```
        $accountsUrl = config('zoho.accounts_url');
```

```
        $response = Http::asForm()->post("{$accountsUrl}/oauth/v2/token", [
            'client_id' => $clientId,
            'client_secret' => $clientSecret,
            'redirect_uri' => $redirectUri,
            'grant_type' => 'authorization_code',
            'code' => $code,
        ];
```

```
    );
```

```

$data = $response->json();

if (isset($data['access_token'])) {
    Session::put('zoho_access_token', $data['access_token']);
    Session::put('zoho_refresh_token', $data['refresh_token']);
    Session::put('zoho_token_expires_in', now()->addSeconds($data['expires_in']));
    return redirect('/dashboard')->with('success', 'Zoho Voice connected successfully');
} else {
    return redirect('/')->with('error', 'Failed to obtain access token: ' . $data['error']);
}
}

public function refreshAccessToken()
{
    $refreshToken = Session::get('zoho_refresh_token');

    if (!$refreshToken) {
        return response()->json(['error' => 'Refresh token not found.'], 404);
    }

    $clientId = config('zoho.client_id');
    $clientSecret = config('zoho.client_secret');
    $accountsUrl = config('zoho.accounts_url');

    $response = Http::asForm()->post("${accountsUrl}/oauth/v2/token", [
        'client_id' => $clientId,
        'client_secret' => $clientSecret,
        'grant_type' => 'refresh_token',
        'refresh_token' => $refreshToken,
    ]);

    $data = $response->json();

    if (isset($data['access_token'])) {
        Session::put('zoho_access_token', $data['access_token']);
        Session::put('zoho_token_expires_in', now()->addSeconds($data['expires_in']));
        return response()->json(['message' => 'Access token refreshed successfully']);
    } else {
        return response()->json(['error' => 'Failed to refresh access token: ' . $data['error']]);
    }
}
}

```

3.3 Define Routes

Add the following routes to your `routes/web.php` file:

PHP

```
<?php

use App\Http\Controllers\ZohoOAuthController;
use Illuminate\Support\Facades\Route;

Route::get('/zoho/oauth/redirect', [ZohoOAuthController::class, 'redirectToZoho']);
Route::get('/zoho/oauth/callback', [ZohoOAuthController::class, 'handleZohoCallback']);
Route::get('/zoho/oauth/refresh', [ZohoOAuthController::class, 'refreshAccessToken']);

// Example dashboard route
Route::get('/dashboard', function () {
    return 'Welcome to your dashboard!';
})->middleware('web');
```

4. Zoho Voice API Service

Create a dedicated service class to encapsulate Zoho Voice API interactions.

4.1 Create ZohoVoiceService Class

Bash

```
php artisan make:class Services/ZohoVoiceService
```

PHP

```
<?php

namespace App\Services;

use Illuminate\Support\Facades\Http;
use Illuminate\Support\Facades\Session;

class ZohoVoiceService
{
    protected $baseUrl;

    public function __construct()
    {
        $this->baseUrl = config('zoho.voice_api_url');
    }

    protected function getAccessToken()
```

```

{
    // Implement logic to check token expiry and refresh if needed
    // For simplicity, this example assumes token is always valid or refreshes
    return Session::get('zoho_access_token');
}

protected function makeRequest($method, $endpoint, $data = [], $headers = [])
{
    $accessToken = $this->getAccessToken();

    if (!$accessToken) {
        throw new \Exception('Zoho access token not available.');
```

```

        'Accept' => 'application/json',
    ]);

    if ($mmsMediaFilePath && $isMms) {
        $request->attach(
            'mms_media',
            file_get_contents($mmsMediaFilePath),
            basename($mmsMediaFilePath)
        )->asMultipart();
    } else {
        $request->asForm(); // For non-multipart requests
    }

    $response = $request->post("{ $this->baseUrl }/{ $endpoint }", [
        'sms_data' => json_encode($smsData),
    ]);

    return $response->json();
}

/**
 * Get SMS logs.
 * Documentation: https://www.zoho.com/voice/api/get-sms-logs.html
 *
 * @param array $params Query parameters for filtering logs.
 * @return array
 */
public function getSmsLogs(array $params = [] )
{
    $endpoint = 'v1/sms/logs';
    return $this->makeRequest('get', $endpoint, $params);
}

/**
 * Get Users.
 * Documentation: https://www.zoho.com/voice/api/user-api.html
 *
 * @param array $params Query parameters for filtering users.
 * @return array
 */
public function getUsers(array $params = [] )
{
    $endpoint = 'v1/users';
    return $this->makeRequest('get', $endpoint, $params);
}

/**
 * Get Call Logs.

```

```

* Documentation: https://www.zoho.com/voice/api/call-logs-api.html
*
* @param array $params Query parameters for filtering call logs.
* @return array
*/
public function getCallLogs(array $params = [] )
{
    $endpoint = 'v1/calllogs';
    return $this->makeRequest('get', $endpoint, $params);
}
}

```

5. Usage Examples

Here's how you can use the `ZohoVoiceService` in your controllers or other parts of your Laravel application.

5.1 Sending an SMS

PHP

```

<?php

namespace App\Http\Controllers;

use App\Services\ZohoVoiceService;
use Illuminate\Http\Request;

class SmsController extends Controller
{
    protected $zohoVoiceService;

    public function __construct(ZohoVoiceService $zohoVoiceService)
    {
        $this->zohoVoiceService = $zohoVoiceService;
    }

    public function sendSms(Request $request)
    {
        $customerNumber = $request->input('customer_number'); // e.g., '1234567
        $message = $request->input('message');
        $senderId = 'your_zoho_voice_number'; // Your Zoho Voice number

        try {
            $response = $this->zohoVoiceService->sendSmsV2($customerNumber, $me
            return response()->json(['status' => 'success', 'data' => $response

```



```

    } catch (\Exception $e) {
        return response()->json(['status' => 'error', 'message' => $e->getMessage()]);
    }
}

public function sendMms(Request $request)
{
    $customerNumber = $request->input('customer_number');
    $message = $request->input('message');
    $senderId = 'your_zoho_voice_number';
    $imagePath = storage_path('app/public/image.jpg'); // Ensure this file exists

    try {
        $response = $this->zohoVoiceService->sendSmsV2($customerNumber, $message, $senderId, $imagePath);
        return response()->json(['status' => 'success', 'data' => $response]);
    } catch (\Exception $e) {
        return response()->json(['status' => 'error', 'message' => $e->getMessage()]);
    }
}
}

```

5.2 Retrieving SMS Logs

PHP

```

<?php

namespace App\Http\Controllers;

use App\Services\ZohoVoiceService;
use Illuminate\Http\Request;

class SmsLogController extends Controller
{
    protected $zohoVoiceService;

    public function __construct(ZohoVoiceService $zohoVoiceService)
    {
        $this->zohoVoiceService = $zohoVoiceService;
    }

    public function getLogs(Request $request)
    {
        $params = [
            'from' => $request->input('from_date'), // Optional: 'yyyy-mm-dd'
            'to' => $request->input('to_date'),      // Optional: 'yyyy-mm-dd'
            'limit' => $request->input('limit', 20),
        ];
    }
}

```

```

        'offset' => $request->input('offset', 0),
    ];

    try {
        $response = $this->zohoVoiceService->getSmsLogs($params);
        return response()->json(['status' => 'success', 'data' => $response]);
    } catch (\Exception $e) {
        return response()->json(['status' => 'error', 'message' => $e->getMessage()]);
    }
}
}

```

5.3 Retrieving Call Logs

PHP

```

<?php

namespace App\Http\Controllers;

use App\Services\ZohoVoiceService;
use Illuminate\Http\Request;

class CallLogController extends Controller
{
    protected $zohoVoiceService;

    public function __construct(ZohoVoiceService $zohoVoiceService)
    {
        $this->zohoVoiceService = $zohoVoiceService;
    }

    public function getCallLogs(Request $request)
    {
        $params = [
            'from' => $request->input('from_date'), // Optional: 'yyyy-mm-dd'
            'to' => $request->input('to_date'),      // Optional: 'yyyy-mm-dd'
            'limit' => $request->input('limit', 20),
            'offset' => $request->input('offset', 0),
        ];

        try {
            $response = $this->zohoVoiceService->getCallLogs($params);
            return response()->json(['status' => 'success', 'data' => $response]);
        } catch (\Exception $e) {
            return response()->json(['status' => 'error', 'message' => $e->getMessage()]);
        }
    }
}

```

```
}  
}
```

6. Important Considerations

- **Token Management:** The example uses `Session` for storing tokens. For a production application, consider a more robust and secure storage mechanism (e.g., database, encrypted cache) and implement a proper token expiry check and automatic refresh mechanism within `getAccessToken()`.
- **Error Handling:** Implement comprehensive error handling and logging for API requests and responses.
- **Environment Variables:** Always use environment variables for sensitive information like Client ID and Client Secret.
- **Scopes:** Ensure you request only the necessary OAuth scopes for your application's functionality.
- **Rate Limits:** Be mindful of Zoho Voice API rate limits to avoid getting blocked.
- **Webhooks:** For real-time updates (e.g., incoming calls, SMS), explore Zoho Voice webhooks to avoid constant polling.

This guide provides a foundational setup. You can extend the `ZohoVoiceService` to include more Zoho Voice API endpoints as needed for your application.

References

1. [Getting Started with Zoho Voice REST APIs](#)
2. [Introduction to OAuth 2.0](#)
3. [POST - Send SMS v2 version](#)
4. [Users API](#)
5. [Call Logs API](#)